

AI Project Manager

Trello + n8n + Gemini + Telegram

An n8n workflow that reads a Trello board, decides project health with clear rules, has Gemini write a short founder report, checks it against the rules, and sends it on Telegram.

• OVERDUE

• BLOCKED

• AT RISK

• ON TRACK

Founders hear about late projects too late

The data is already in Trello, but nobody opens every card every morning.

Scattered signals

Deadlines, hours, blockers and dependencies sit in different fields on different cards.

Hidden knock-on effects

One blocked card quietly delays every card that depends on it.

AI alone is not reliable

Asking a model "is this at risk?" can give a different answer each run, with no audit trail.

Rules decide. AI explains. n8n delivers.

Trello stays the source of truth

No new tool for the team. Lists, due dates and 9 custom fields.

n8n runs it every weekday

08:00 schedule plus a manual run button. 10 nodes, importable JSON.

Deterministic rule engine

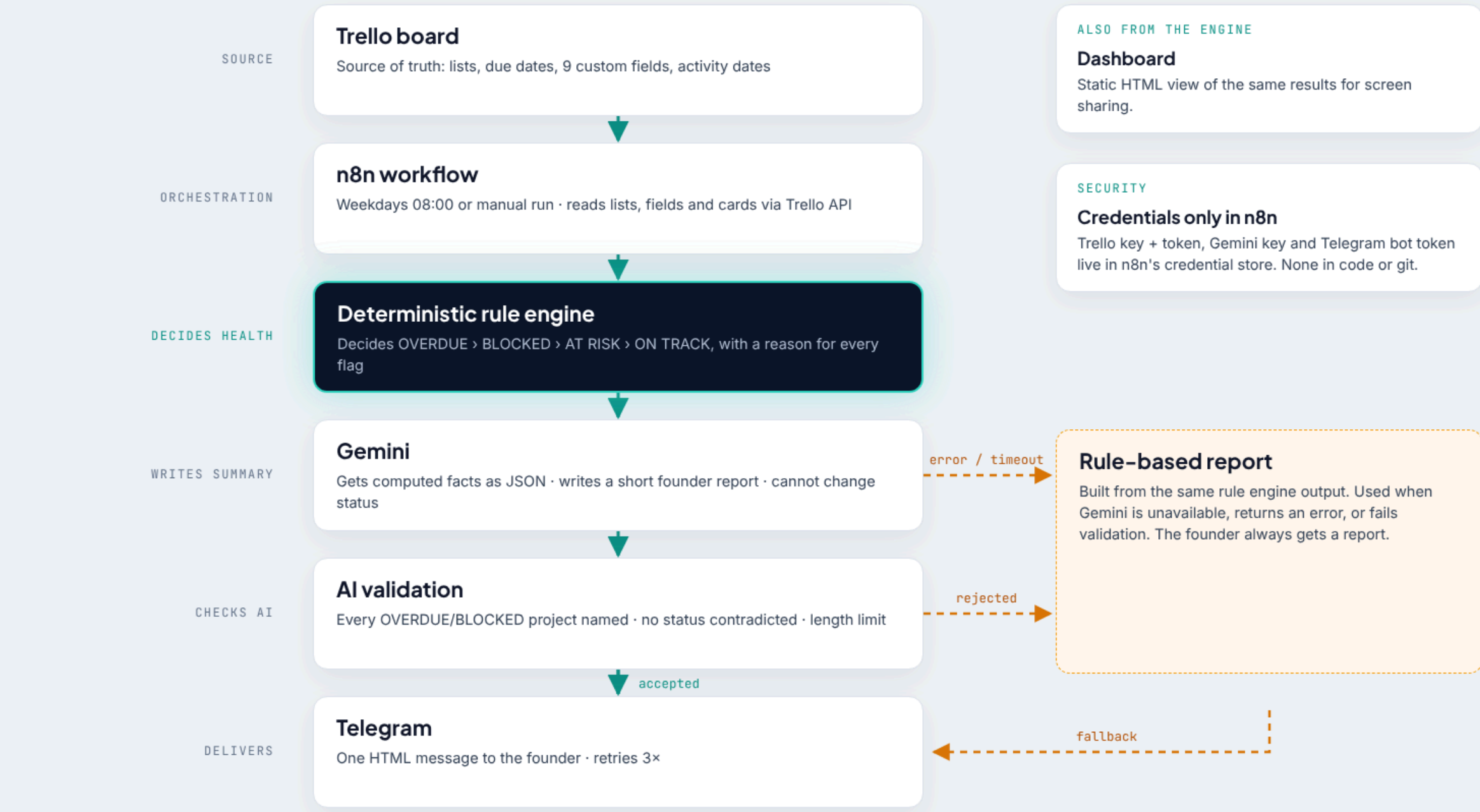
Same input, same answer. Every flag carries a plain-English reason.

Gemini writes, rules check

AI text is used only if it agrees with the rules. Otherwise the rule report is sent.

Rules decide. AI explains. n8n delivers.

Main path on the left. If Gemini fails, times out, or writes something that contradicts the rules, the rule-based report goes to Telegram instead.



The board the workflow reads

12 projects · 6 lists · owner, priority, deadline, hours, blocker, dependency, risk level, next milestone, decision required.

Trello board

Rendered from Trello REST API data · board b0800000... · as of 2026-10-07

P08 Delivery Portfolio

BoardCustom Fields: 912 cards

Backlog1

Medium

Knowledge Base Cleanup

🕒 10 Nov

🕒 0/20h

RiskLow

NextArticle audit

HY

To Do1

Critical

Security Audit Remediation

🕒 12 Oct

🕒 0/24h

RiskMedium

NextFix high-severity findings

DecisionApprove the external pen-test vendor

KR

In Progress6

Critical

Client Portal v2

🕒 2 Oct

🕒 138/120h

RiskHigh

NextUAT sign-off

DecisionApprove a 1-week scope cut or one extra developer

SM

High

Mobile App Release 3.4

🕒 18 Oct

🕒 52/80h

Depends onPayment Gateway Migration

RiskMedium

NextApp Store submission

LF

Medium

Data Warehouse Sync

🕒 25 Oct

🕒 55/40h

RiskMedium

NextNightly sync in staging

BQ

Medium

Marketing Website Refresh

🕒 6 Nov

🕒 22/50h

RiskLow

NextHomepage design review

NS

Blocked1

High

Payment Gateway Migration

🕒 20 Oct

🕒 41/60h

BlockerWaiting for bank sandbox credentials

RiskMedium

NextFirst sandbox transaction

OH

Review2

Low

Customer Onboarding Emails

🕒 28 Oct

🕒 9/16h

RiskLow

NextCopy approval

HY

Medium

CRM Data Import

🕒 21 Oct

🕒 17/18h

RiskLow

NextData validation sign-off

BQ

Done1

Medium

Q3 Invoicing Automation

🕒 30 Sept

🕒 33/35h

RiskLow

OH

Captured while it runs

Real execution in n8n: 4.3 s from click to finish. Each node turns green with its item count.

Top: 2.6 s after clicking Execute — Trello and Risk Engine done, Gemini running. Bottom: every node green, item counts on each connection.

● RUNNING

● SUCCEEDED

● 34/34 TESTS

n8n • executing • t = 2.6 s

● RUNNING

Run Now

Weekdays 08:00

Config manual

Trello: Get Lists

Trello: Get Custom Fields

Trello: Get Cards

Risk Engine

Gemini: Founder Report

Select Report

Telegram: Notify Founder

1 Item

6 Items

9 Items

12 Items

1 Item

sendMessage: message

n8n • workflow executed successfully

● SUCCEEDED

Nine signals, fixed precedence

SIGNAL	RULE	SEVERITY
• OVERDUE	Deadline before today, card not in Done	critical
• BLOCKED	Blocker field filled, or card in the Blocked list	critical
• DUE SOON, NOT STARTED	Deadline within 7 days, still in Backlog / To Do	high
• DEPENDENCY AT RISK	Depends on a card that is OVERDUE or BLOCKED	high
• DEPENDENCY LATE / NOT FOUND	Dependency finishes after this deadline, or name not on the board	high
• HOURS OVER ESTIMATE	Actual > estimated hours (high above +20%)	medium / high
• MISSING UPDATE	No card activity for more than 7 days (high above 14)	medium / high
• MISSING REQUIRED DATA	Owner, Deadline, Priority or Estimated Hours empty	medium
• OWNER REPORTED HIGH RISK	Risk Level custom field = High	medium

The AI gets facts, not decisions

Gemini receives the computed results as JSON with strict instructions: never change a status, use only the given facts, name every overdue and blocked project, max 170 words.

AI REPORT FROM THE RUN (ACCEPTED)

2 projects need you today; 6 more are at risk.

Act now

- Client Portal v2 (Sara Malik) is overdue: Deadline 2026-10-02 passed 5 day(s) ago.
- Payment Gateway Migration (Omar Haddad) is blocked: Blocker: Waiting for bank sandbox credentials.

Watch

- Security Audit Remediation: Due in 5 day(s) but still in "To Do".
- Mobile App Release 3.4: Depends on "Payment Gateway Migration", which is BLOCKED.
- Support Chatbot Rollout: Depends on "Knowledge Base Cleanup" (due 2026-11-10),

which finishes after this deadline (2026-10-30).

- Data Warehouse Sync: 55h logged vs 40h estimated (+38%).
- Customer Onboarding Emails: No update for 15 days.
- Internal Analytics Dashboard: Missing: Owner.

Decisions for you

- Client Portal v2: Approve a 1-week scope cut or one extra developer.
- Security Audit Remediation: Approve the external pen-test vendor.

On track: CRM Data Import, Knowledge Base Cleanup, Marketing Website Refresh.

The founder always gets a correct report

Correct report

• AI REPORT SENT

AI report passed validation

Telegram ok • message #13

Contradicts rules

• RULE REPORT SENT

AI report rejected: AI response calls "Security Audit Remediation" on track, rules say AT RISK

Telegram ok • message #14

Gemini unavailable

• RULE REPORT SENT

AI request failed: HTTP 503 — model unavailable

Telegram ok • message #15

Checks on the AI text

Not empty, under 2,500 characters • names every OVERDUE and BLOCKED project • never describes a non-healthy project as "on track".

One clear message, every weekday 08:00

Sent by the n8n Telegram node. The Telegram Bot API answered ok = true (right: the Bot API response from the recorded editor run). Chat and bot IDs are masked.

• DELIVERED • HTML FORMAT • RETRY 3*

P08 Project Health — 2026-10-07

● Overdue 1 ● Blocked 1 ● At risk 6 ● On track 3

Status decided by rules · report by Gemini-compatible test endpoint (validated)

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n8n • Telegram node • Bot API response

OUTPUT

SchemaTableJSON

This node will automatically retry if it fails

1 item

ok	result
true	message_id : 16 from id : is_bot : true first_name : AI Project Manager username : rehan_ai_project_manager_bot chat id : first_name : Rehan last_name : khan type : private

12 scenarios - 34/34 automated tests pass

#	SCENARIO	EXPECTED AND VERIFIED	WHERE
S1	All projects on track	No flags, no "act now" section	Unit
S2	Overdue project	OVERDUE, days late in reason	Unit + n8n
S3	Blocked project	BLOCKED, blocker text in reason	Unit + n8n
S4	Dependency problem	Blocked / late / unknown dependency flagged	Unit + n8n
S5	Hours exceeded	+38% → high severity	Unit + n8n
S6	Multiple risks on one card	6 flags, OVERDUE wins by precedence	Unit
S7	Missing update	15 days → high severity	Unit + n8n
S8	Missing required data	Missing owner listed	Unit + n8n
S9	Gemini unavailable	HTTP 503 → rule report sent	Unit + n8n
S10	Gemini correct	AI report accepted	Unit + n8n
S11	Gemini contradicts rules	Rejected → rule report sent	Unit + n8n
S12	Telegram notification	Delivered, Bot API ok = true	Unit + n8n

What the system does today

Same workflow, same Trello data, three Gemini behaviours. Every run reached Telegram. Plus 34/34 automated tests on the engine.

- 34/34 TESTS PASS
- 3/3 n8n RUNS DELIVERED
- 12/12 SCENARIOS

GEMINI BEHAVIOUR	REPORT SENT	SELECT REPORT REASON	TELEGRAM	NODES OK
Correct report p08-test-faithful	● AI REPORT	AI report passed validation	● ok · #13	9/9
Contradicts rules p08-test-contradict	● RULE REPORT	AI report rejected: AI response calls "Security Audit Remediation" on track, rules say AT RISK	● ok · #14	9/9
Unavailable (503) p08-test-unavailable	● RULE REPORT	AI request failed: HTTP 503 — model unavailable	● ok · #15	9/9

ACCEPTANCE SCENARIOS

- ✓ S1 All projects on track

No flags, no "act now" section
- ✓ S2 Overdue project

OVERDUE, days late in reason
- ✓ S3 Blocked project

BLOCKED, blocker text in reason
- ✓ S4 Dependency problem
- ✓ S5 Hours exceeded
- ✓ S6 Multiple risks on one card

Small, standard and easy to hand over

n8n 1.123

Schedule + manual triggers, HTTP Request, Code, Telegram nodes. Credentials stored in n8n.

Trello REST API

Lists, custom fields and open cards. Key + token via the n8n Trello credential.

Google Gemini API

generateContent, temperature 0.2, 30 s timeout, error output continues the flow.

JavaScript rule engine

One dependency-free file, embedded unchanged in the n8n Code nodes.

Telegram Bot API

HTML message, link previews off, 3 retries.

Node.js test runner

34 tests: rules, edge cases, invalid input, parsing, validation, message format.